

Calum Milloy

Data Analyst

PhD Physicist and Data Analyst with 6+ years' experience developing Python-based analytical software, machine learning models and statistical simulations in research environments. Experienced analysing large datasets, building predictive models and communicating complex technical findings. Currently teaching mathematics while transitioning into industry.

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SKILLS

Analytical
Problem-Solving

Independent Work
Python

SQL
NLP

AI tools
Communication

EXPERIENCE

Mathematics Teacher

Vittoria International School 2023 – Present Turin, Italy

- Teach IB and IGCSE Mathematics, communicating effectively to classes of 25+ students.
- Use AI tools to develop interactive learning resources and streamline assessment and feedback processes.
- Analyse student performance data to identify learning gaps and adapt teaching strategies to improve outcomes.
- Manage classes of 25+ students while delivering planned lessons and adapting effectively to changing classroom needs.

Postdoctoral Researcher

Università di Torino 2020 – 2023 Turin, Italy

- Optimized massive numerical datasets (10^6 + data points), reducing computational overhead by implementing machine learning algorithms using scikit-learn to predict analytical results.
- Developed efficient scripts in C++, Python and Mathematica to handle the computational load, ensuring precision and stability in large-scale calculations.
- Collaborated with international colleagues to drive research projects to completion under tight deadlines, using Git to share a common codebase and manage version control.
- Published 7 peer-reviewed papers in high-impact journals and delivered research talks at 6 international conferences.
- Elected representative to the Consiglio di Dipartimento; collaborated with department leadership to manage resource allocation and represent the interests of the researchers.

EDUCATION

PhD Physics

University of Edinburgh 2016 – 2020

- Research focused in fundamental theoretical physics.
- Thesis title: *Infrared singularities of scattering amplitudes in perturbative QCD*
- Relevant Courses: C++, Python, Advanced Data Analysis

MPhys (Hons) Mathematical Physics

University of Edinburgh 2012 – 2016

- First Class
- Relevant Courses: Advanced Statistical Physics, Methods of Mathematical Physics, Programming & Data Analysis, Computer Modelling

SELECTED DATA SCIENCE PROJECTS

Predictive Maintenance Platform

- Designed and implemented a PostgreSQL database from five relational manufacturing datasets, using SQL and pandas to support exploratory analysis and feature engineering.
- Developed an end-to-end machine learning pipeline to predict equipment failures 7 days in advance from telemetry, maintenance and historical failure data.
- Built an interactive Streamlit application that combines failure predictions with SHAP-based explanations to support maintenance decision-making.

Visualise HEP

- Developed an end-to-end data pipeline using Python (spaCy, SentenceTransformers) to extract and process natural language data from arXiv API to identify emerging physics research fields.
- Designed an interactive Plotly Dash dashboard featuring a WordCloud of the emerging terms.

CERTIFICATION & COURSES

Databricks, AI Agent and Generative AI Fundamentals
Databricks

IBM Data Science Professional Certificate
Coursera

AWS Cloud Practitioner Essentials
AWS

Python
freeCodeCamp

AWARDS

Winton Capital Thesis Prize
for best PhD thesis in Nuclear & Particle Physics

Tait Medal and Robin Schlapp Prize
for best result in Mathematical Physics degree

Carnegie PhD Scholarship
a prestigious award for PhD funding

Margaret Ann Stewart Memorial Prize
for highest overall grade across all degree programmes in the School of Physics & Astronomy

LANGUAGES

English
Mother tongue

Italian
B1

Spanish
A1